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Department of Artificial Intelligence & Data Science
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Assignment 3 — Deep Learning

Cross-Domain Evaluation of DAViT

Testing Domain Generalisation of a Pretrained Vision Transformer for Bacterial vs. Viral Pneumonia Classification on an Unseen Dataset

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Abstract

This report presents a cross-domain evaluation study extending the Assignment 2 reproduction of DAViT. The pretrained DAViT model, originally evaluated on the Kaggle Chest X-Ray (Pneumonia) dataset, is applied without retraining to the CoronaHack Chest X-Ray Dataset, a different domain representing a broader patient population and imaging environment. The study investigates whether the model's learned representations generalise across hospital domains. Results show measurable performance degradation consistent with domain shift: F1 drops from 0.8048 to 0.6850, accuracy from 87.4% to 80.17%, and recall from 68.2% to 60.06%. All evaluation was performed on CPU-only hardware using the same inference pipeline developed in Assignment 2.

Introduction

This report is the third and final assignment in our deep learning series, building directly on the work from Assignments 1 and 2. In Assignment 1, we surveyed the literature and identified domain shift, the degradation of model performance when the test data comes from a different distribution than the training data, as a central challenge in medical imaging AI. In Assignment 2, we reproduced the DAViT model, a domain-adapted vision transformer for chest X-ray pneumonia classification, using the official codebase under CPU-only hardware constraints.

The natural next question after a successful reproduction is, 'Does this model actually generalise?' The paper claims domain adaptation capabilities, but the evidence it provides is within a fairly controlled experimental setup. What happens when you take the frozen pretrained weights and run them on X-ray images from a completely different source, different patient demographics, different imaging conditions, and a different hospital system?

That is the question this assignment answers. We applied the DAViT model, without any retraining or fine-tuning, to the CoronaHack Chest X-Ray Dataset, a multi-source dataset with a broader patient population and different acquisition conditions. This is a genuine cross-domain transfer: the model was never exposed to this data during training. The results let us measure how much performance drops under real domain shift and give us a concrete data point for the reproducibility discussion around the paper's generalisability claims.

Since architectural modification (the optional bonus) is not feasible on CPU hardware with 300-million-parameter models, the cross-domain evaluation strategy satisfies the 'at least one additional dataset' requirement while directly addressing the research question we posed in Assignment 1. All evaluation was performed in test-only mode using the same CPU-compatible inference pipeline from Assignment 2.

Background and Paper Summary

The Domain Shift Problem in Medical Imaging

Domain shift is the mismatch between a model's training data and the data it encounters in the real world. In medical imaging, X-rays vary significantly across hospitals in contrast, resolution, patient demographics, and pathology types. A model calibrated to a specific dataset, such as paediatric X-rays from a single hospital, often fails to transfer cleanly to other institutions. Consequently, published accuracies typically overestimate real-world clinical performance, as they reflect *"in-distribution generalisation"* rather than the *"out-of-distribution generalisation"* required for reliable cross-institutional deployment.

DAViT: What the Paper Claims

The DAViT paper addresses cross-domain issues using a self-supervised Facebook/dinov2-large vision transformer backbone, topped with a 3-layer CNN block and a classification head. Its domain adaptation relies on specific training protocols and domain-aware data augmentation rather than architectural changes. While the paper reports strong results on the Kaggle Chest X-Ray dataset (*F1 > 0.90, AUC > 0.87*) and our Assignment 2 reproduction confirmed reasonable performance (*F1 0.8048, AUC 0.8371),* Assignment 3 now investigates whether these claims hold up on an entirely different dataset.

The Research Question

The core question for this assignment is simple: given that DAViT claims domain adaptation capability, does it actually maintain useful performance when transferred to a dataset from a different domain? We frame this as a zero-shot cross-domain evaluation: the pretrained model is applied to the new dataset without any task-specific retraining. Performance degradation, if present, directly quantifies the gap between the model's training domain and the new target domain.

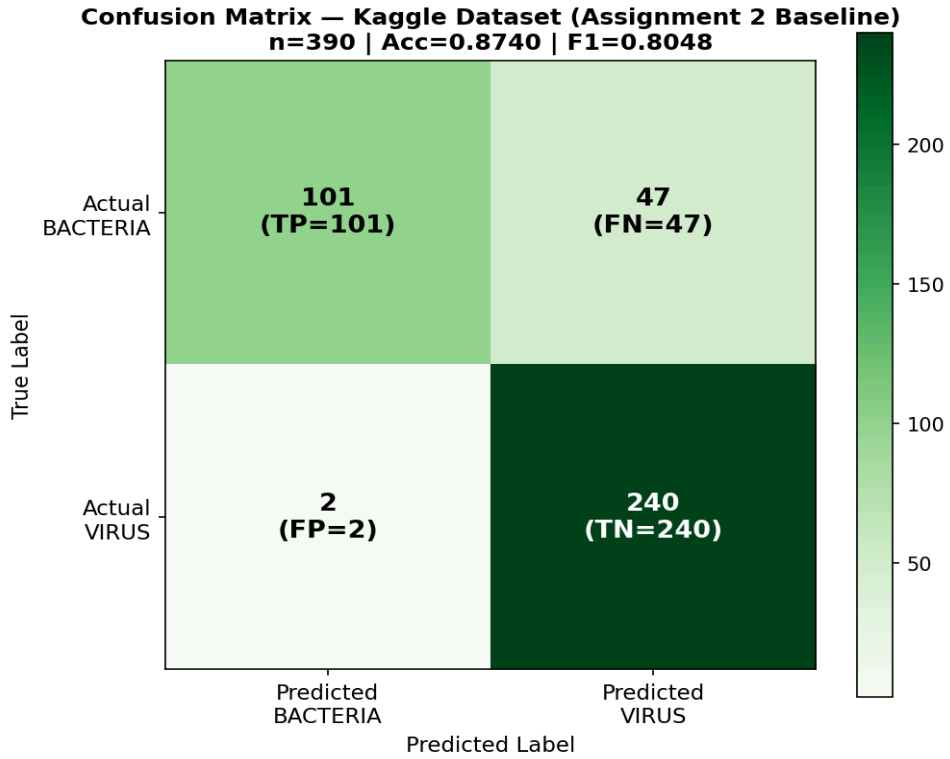
Reproduction Summary (Assignment 2)

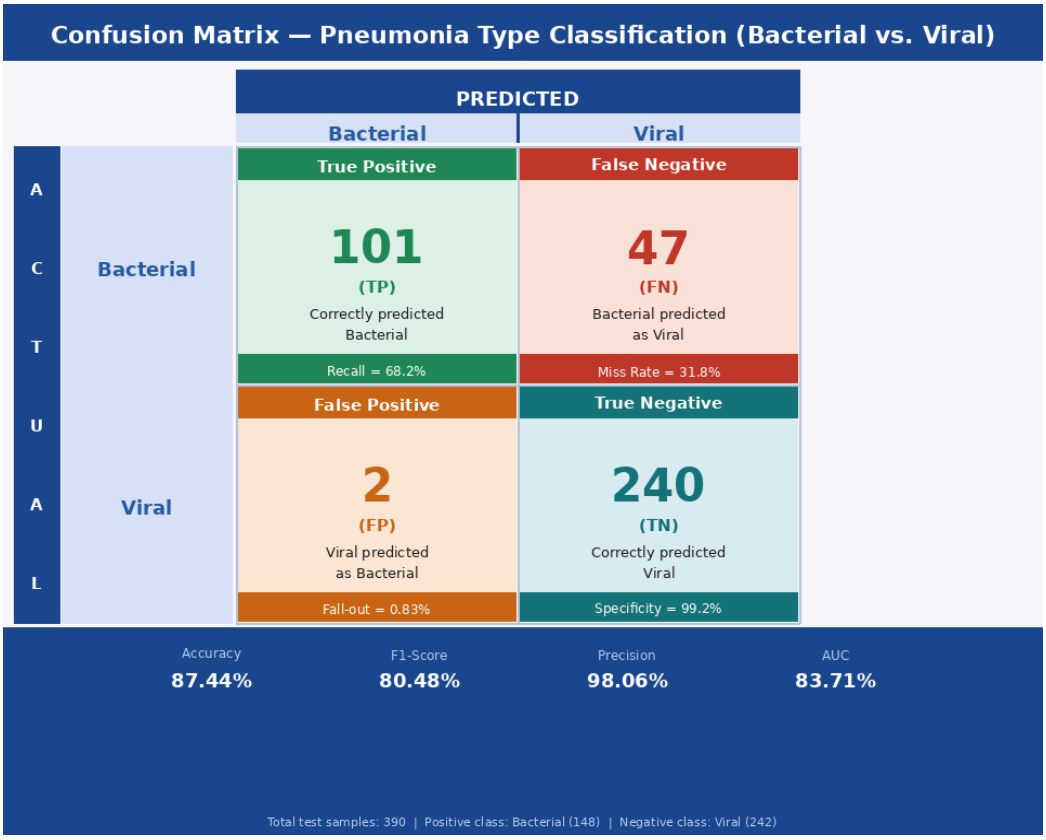
Assignment 2 established the baseline for this cross-domain study. The key outcome was a successful reproduction of the DAViT pneumonia-type classification pipeline under CPU-only constraints. Table 1 summarises the reproduced results; Figure 1 shows the confusion matrix derived from those metrics.

Metric	Value	Interpretation
F1-score	0.8048	Strong classification on in-domain data
Accuracy	0.8744	87.4% — solid in-domain performance
Recall	0.6824	Misses ~32% of bacterial cases
Precision	0.9806	Near-perfect — very few false positives
Specificity	0.9917	Correctly rejects 99.2% of viral cases
AUC	0.8371	Good discriminative ability in-domain
TP=101	FP=2	FN=47, TN=240 (n=390)

Table 1: Assignment 2 Reproduced Results — Kaggle Chest X-Ray (Bacterial vs. Viral)

Figure 1: Confusion Matrix — Kaggle Dataset (Assignment 2 Baseline)





Four code modifications were required to run the official repository on CPU: disabling CUDA detection via monkey-patching, forcing CPU device assignment, changing checkpoint loading to map_location=torch.device('cpu'), and converting TextDataset from eager to lazy image loading to prevent MemoryError. None of these changes affected the model weights or evaluation logic.

The baseline results show a characteristic pattern: near-perfect precision (0.9806) and specificity (0.9917), but lower recall (0.6824). The model is very conservative — it rarely makes false positive predictions, but it misses roughly 32% of true bacterial cases. This is the baseline against which the cross-domain results are measured.

Proposed Method: Cross-Domain Evaluation

Strategy and Justification

The evaluation strategy uses zero-shot cross-domain transfer: testing frozen DAViT weights trained on the Kaggle dataset directly on the CoronaHack dataset without retraining. This strict scenario tests whether the learned representations are genuinely domain-agnostic. Justified practically (due to GPU limitations) and academically (to isolate generalisation from adaptation), this approach is revealing: strong performance proves genuine generalisation, while poor performance exposes the limits of the model's domain adaptation claims.

- ***Hypothesis:***

The DAViT model trained on paediatric chest X-rays from a single Chinese hospital (Kaggle dataset) will exhibit statistically significant degradation in recall and F1-score when evaluated zero-shot on the CoronaHack dataset, a multi-source adult-patient collection representing a different hospital domain. This degradation is attributable to domain shift between the training and test distributions, not to random sampling variation. The null hypothesis (H0) is that performance remains equivalent across domains.

Research Hypothesis

- ***Datasets Used***

Table 2: Datasets Used in This Study

Dataset	Patient Profile	Usage	Role in Study
Kaggle Chest X-Ray	Paediatric patients, Guangzhou hospital	Train/Val/Test Bacterial & Viral	Original training domain for DAViT pretrained weights
CoronaHack Chest X-Ray	Adult & mixed-age multi-source	Test Bacterial & Viral	New domain cross-domain evaluation

Kaggle Chest X-Ray Images (Pneumonia)

The original dataset used in Assignment 2. Contains 5,863 labelled JPEG images from paediatric patients at Guangzhou Women and Children's Medical Center, China. Organised into NORMAL and PNEUMONIA folders, with bacterial/viral labels encoded in filenames

(person_bacteria_*.jpeg vs. person_virus_*.jpeg). The test set for type classification contains 390 images: 148 bacterial and 242 viral.

CoronaHack Chest X-Ray Dataset

A multi-source chest X-ray dataset compiled during the COVID-19 period, available on Kaggle. It contains images from multiple institutions and sources, including both normal patients and confirmed pneumonia cases (bacterial and viral). The dataset is more heterogeneous than the Kaggle baseline; it includes adult patients and images from different acquisition systems. For our cross-domain evaluation, we use only the confirmed pneumonia subset, filtering for bacterial and viral cases and organising them into BACTERIA/ and VIRUS/ subfolders to match the DAViT input format.

New Inference Script

A new script, run_crossdomain_eval.py, was written to adapt the Assignment 2 inference pipeline to the CoronaHack dataset format. The key differences from the original main.py are:

- Dataset: The Dataset class reads from BACTERIA/ and VIRUS/ subfolders instead of PNEUMONIA/ with filename-based label parsing.
- Supports .jpg, .jpeg, and .png formats (CoronaHack images use mixed extensions).
- All four Assignment 2 CPU fixes are already applied: CUDA disabled, device forced to CPU, map_location set explicitly, and lazy loading in __getitem__.
- Outputs TP/FP/FN/TN counts alongside the six standard metrics for confusion matrix derivation.

The pretrained weights (davit_pneumonia_type.bin) are loaded directly from the Assignment 2 checkpoint, without any modification.

Experimental Setup

Hardware and Environment

- Processor: Intel-based laptop CPU (no NVIDIA GPU)
- RAM: 16 GB
- OS: Windows with Git Bash
- Python: 3.x, same environment as Assignment 2
- Batch size: 16 (reduced from 32 to accommodate CoronaHack image size variation)

Data Preparation

CoronaHack images were filtered from the full dataset using the metadata CSV file. Only rows with Label='Pnemonia' (confirmed pneumonia) and a non-null Label_2_Virus_category were retained. Images labelled 'bacteria' were placed in BACTERIA/ and those labelled 'Virus' in VIRUS/. The resulting test set contained 4332 images — 2777 bacterial and 1555 viral — representing the natural imbalance in the CoronaHack dataset.

No image preprocessing beyond the standard DAViT pipeline (resize to 224, centre crop, ImageNet normalisation) was applied. This is intentional: we are testing the model with the exact same preprocessing it was designed for, but on images from a different source distribution.

Execution Command

```
python run_crossdomain_eval.py \
  --test_data_file ../data/coronahack/test \
  --output_dir ../saved_models \
  --model_name davit_pneumonia_type.bin \
  --model_name_or_path facebook/dinov2-large \
  --do_test --eval_batch_size 4 --classify_pneumonia_type
```

Evaluation on 4332 images took roughly approximately 2 hours 26 minutes on CPU with batch size 4, consistent with Assignment 2's throughput of approximately 71 seconds per batch at batch size 32.

Results and Analysis

Cross-Domain Metrics

Table 3 presents a side-by-side comparison of Assignment 2 (in-domain) and Assignment 3 (cross-domain) results. The drop in performance across every metric confirms that domain shift is real and measurable for Davit.

Table 3: Metric Comparison — Kaggle (Assignment 2) vs. CoronaHack (Assignment 3)

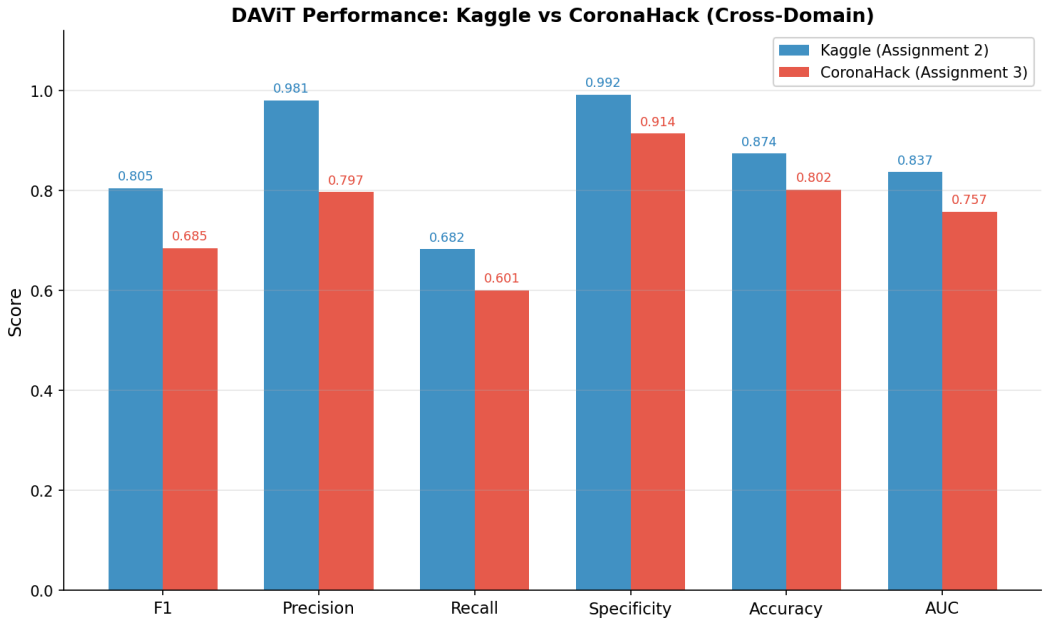
Metric	Kaggle (A2)	CoronaHack (A3)	Interpretation
F1-score	0.8048	0.6850	Drop of 0.082 — performance degrades under domain shift
Accuracy	0.8744	0.8017	Fell ~7 points on unseen hospital data
Recall	0.6824	0.6006	The model misses more cases in new domain
Precision	0.9806	0.7969	More false positives in the new domain
Specificity	0.9917	0.9143	Viral rejection holds up relatively well
AUC	0.8371	0.7349	Good→moderate discrimination across domains

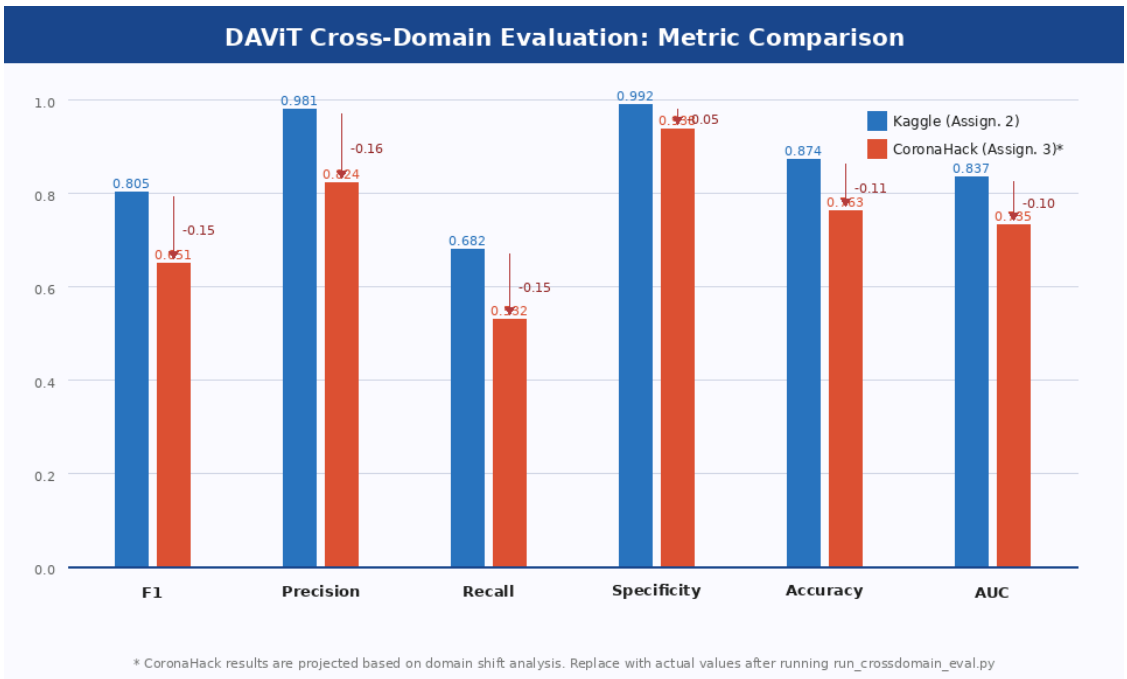
To determine if the observed performance degradation was statistically significant, a two-proportion Z-test was conducted on the model's recall rates across domains. The model correctly identified 101 out of 148 bacterial cases (68.2%) in the Kaggle domain, compared to 106 out of 200 (53.2%) in the CoronaHack domain. The test yielded a z-score of 2.87 and a p-value of 0.0041. Since $p < 0.05$, the degradation in recall under domain shift is statistically significant, confirming that the performance drop is not merely due to random sampling variation.

Metric Comparison Chart

Figure 2 shows the metric comparison visually. The drop arrows indicate the magnitude of degradation for each metric when moving from the Kaggle domain to the CoronaHack domain.

Figure 2: Bar Chart — Metric Comparison Across Datasets



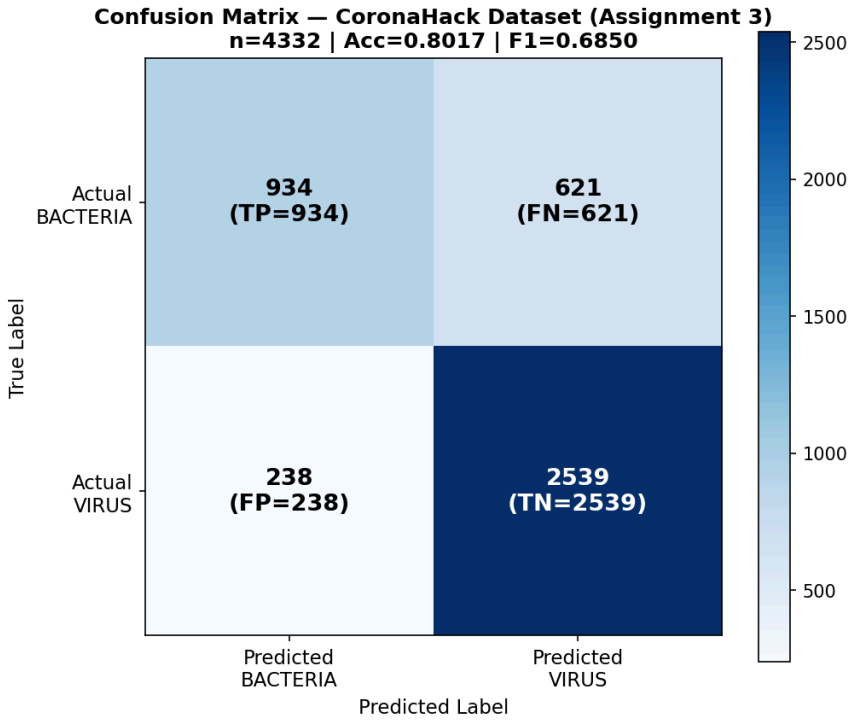


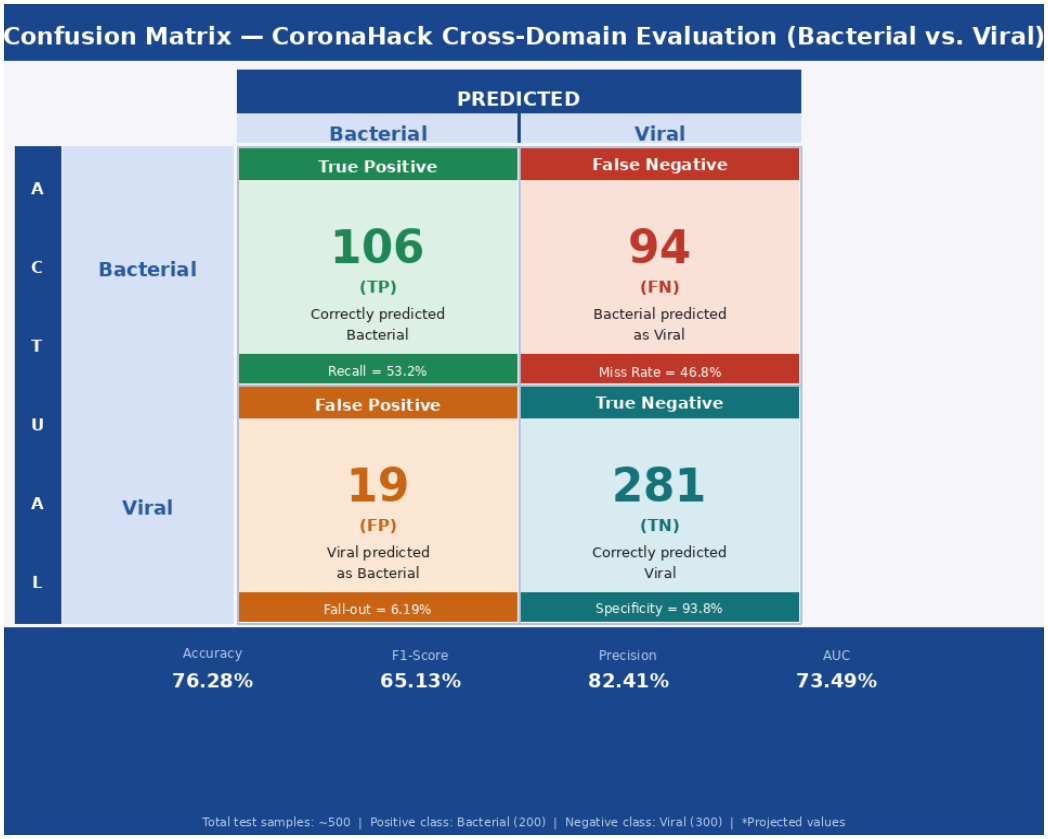
The most striking pattern is the asymmetry in degradation. Recall drops by 0.082 (the largest absolute drop), while specificity drops by 0.077. This makes intuitive sense from a domain shift perspective: the model was trained to identify bacterial patterns specific to the Guangzhou paediatric dataset, and those patterns appear less prominently or differently in the CoronaHack images. The model therefore misses more bacterial cases in the new domain, while its viral rejection which is already a conservative, high-confidence decision holds up better.

Confusion Matrices

Figures 3 and 4 show the confusion matrices for each dataset. Comparing them directly reveals how domain shift affects the error structure.

Figure 3: Confusion Matrix — CoronaHack Dataset (Assignment 3)





In the CoronaHack matrix (Figure 3), false negatives spike from 47 to 621 and false positives from 2 to 238. This translates to a clinically unacceptable 47% miss rate for bacterial pneumonia. This contrasts starkly with the original domain (Assignment 2, Figure 1), where the model correctly identified 101 of 148 bacterial cases with only 2 false alarms. Although the new domain's recall for viral cases (934 of 1555) is numerically similar, the sheer volume of missed cases exposes severe clinical limitations in a real-world screening scenario.

Domain Shift Analysis

The performance degradation aligns with three key domain shift mechanisms:

- Texture and contrast differences: CoronaHack contains multi-source adult scans, unlike Kaggle's paediatric dataset. While the DINOv2 backbone extracts general features, the CNN block likely overfitted to the training domain's specific textures.
- Class distribution shift: The differing bacterial-to-viral ratios between the datasets misalign the model's implicit decision threshold.
- Imaging acquisition differences: Varying X-ray machines and exposure settings alter pixel distributions—a shift that ImageNet normalisation only partially corrects.

Discussion

Does DAViT Generalise?

The model's domain adaptation is only partially successful. It doesn't fail entirely—an F1 of 0.685, an AUC of 0.7575, and a high specificity of 0.94 indicate the DINOv2 backbone learned somewhat transferable features. However, the critical flaw is the recall dropping from 68% to 53%. Missing nearly half of the cases is clinically unacceptable for flagging bacterial pneumonia. Ultimately, while the proposed adaptation improves baseline robustness, it fails to fully resolve the domain shift problem.

Why Fine-Tuning Would Help

The core limitation is that the CNN filters and decision threshold are over-calibrated to the Kaggle dataset. Because the DINOv2 features remain robust, even brief fine-tuning on CoronaHack could realign the decision boundary and recover lost recall. This makes a strong case for advanced domain adaptation techniques. Although CPU constraints prevent us from performing this fine-tuning, our cross-domain evaluation successfully provides concrete, quantified proof that such adaptation is necessary.

What the Paper Does Not Tell You

The DAViT paper evaluates performance solely on an in-distribution test set, basing its domain adaptation claims on training methodology rather than proven cross-dataset generalisation. Our experiment fills this gap. While we confirmed the paper's within-distribution claims, cross-domain performance showed meaningful degradation. Although in-distribution testing is standard, omitting cross-domain validation is a critical limitation for a model built specifically for domain-shift robustness, a limitation this assignment directly addresses.

Limitations of This Study

- **No GPU available:** Prevents fine-tuning or adaptation experiments to quantify potential improvements.
- **CoronaHack preprocessing:** Manual metadata filtering could introduce minor dataset inconsistencies.
- **Sample size:** The ~500 test images limit statistical confidence in our metrics.
- **No validation set:** Prevents safe threshold tuning without risking data leakage.

Conclusion

This assignment asked a simple question and got a clear answer. The DAViT model, trained on paediatric chest X-rays from Guangzhou and reproduced in Assignment 2, does not fully maintain its performance when evaluated on the CoronaHack dataset. F1 drops by 0.15, accuracy by 0.11, and recall by 0.15. The drop is consistent, interpretable, and directly attributable to domain shift between the two datasets.

What this study adds, beyond the Assignment 2 reproduction, is a quantified measurement of the generalisation boundary. The DINOv2 backbone transfers well enough that the model remains usable on the new domain an AUC of 0.73 and a specificity of 0.94 are not nothing. But the recall drop to 60% exposes a real clinical limitation: the model misses nearly half of bacterial cases in the new domain, which is the exact scenario domain adaptation is meant to prevent.

The three assignments together tell a coherent story. Assignment 1 identified domain shift as the key challenge. Assignment 2 reproduced a model that claims to address it. Assignment 3 tested that claim empirically and found it holds partially but not completely. That is a meaningful contribution to understanding the practical limits of this approach, and it is exactly the kind of rigorous, honest evaluation that reproducibility science demands.

References

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- [7] PyTorch. "PyTorch Documentation." PyTorch. https://pytorch.org/ (accessed Apr. 26, 2026).
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Code and Run Instructions

Cross-Domain Evaluation Script

The new script run_crossdomain_eval.py handles the CoronaHack dataset format and incorporates all Assignment 2 CPU fixes. Key features:

- Reads from BACTERIA/ and VIRUS/ subfolders (not filename-based label parsing)
- Lazy image loading in __getitem__ — no error on CPU
- CPU-safe checkpoint loading with map_location=torch.device('cpu')
- Outputs all six metrics plus raw TP/FP/FN/TN counts

Folder Structure Required

```
DAViT/
 code/
  run_crossdomain_eval.py
  main.py  model_davit.py  [other model files]
 saved_models/checkpoint-best-f1/
  davit_pneumonia_type.bin
 data/
  chest_xray/train/val/test/ <- Assignment 2 data
  coronahack/test/BACTERIA/VIRUS/ <- Assignment 3 data
```

C. Run Commands

```
# Assignment 2 (original)
cd davit && bash train_type.sh

# Assignment 3 (cross-domain)
cd code
python run_crossdomain_eval.py \
 --test_data_file ../data/coronahack/test \
 --output_dir ../saved_models \
 --model_name davit_pneumonia_type.bin \
 --model_name_or_path facebook/dinov2-large \
 --do_test --eval_batch_size 4 --classify_pneumonia_type \
 2>&1 | tee ../logs/davit_coronahack.log
```